Problem 1: Nanti (20 marks)

Here are some constructions in Nanti.[1]

igemarite	his tapir
nobonkiti	my foot
iserine	his tobacco
oshiropitonkite	her needle
nogusine	my piping guan
pirirana	your blood
nobarotote	my raft
pibirane	your flashlight
pigotsirote	your knife
igabeharite	his bird-trap
ogito	her head
ibanko	his house
ogobitite	her pot
nogomaginarote	my woolly monkey
ogetane	her walnut
pitsitsihenkate	your soot

Assignment 1: (6 marks)

Translate from Nanti:

(a) kemari

(b) ririrana

(c) ponkiti

(d) pibanko

(e) ogotsirote

(f) noserine

Given below are some additional Nanti nouns.[2]

kuri	peach palm
pori	leg
saburi	machete

[1]The Nanti language is spoken by ∼250 people in Peru. The tapir is a large, pig-like animal. The piping guan is a South American bird species.

[2]The peach palm is a species of palm tree native to Central and South America.

Assignment 2: (6 marks)

Translate to Nanti:

(a) your leg

(b) my peach-palm

(c) his machete

(d) raft

(e) woolly monkey

(f) her piping guan

Assignment 3: (8 marks)

Explain your solution.

– Antara Raaghavi Bhattacharya

Problem 2: Romanian (25 marks)

Here are some phrases in Romanian[3] and their English translations:[4][5]

copilul micuţ al cetăţeanului viteaz	the small child$_M$ of the brave citizen$_M$
aptul cetăţean al nepoatei moarte	the skilled citizen$_M$ of the dead niece
camarazii curioşi ai sătulelor cetăţene	the curious comrades$_M$ of the satiated citizens$_F$
viclenii nepoţi ai înalţilor copii	the cunning nephews of the tall children$_M$
camarada smeadă a mireselor flămânde	the pale comrade$_F$ of the hungry brides
nepoata vicleană a curiosului urs	the cunning niece of the curious bear$_M$
frumosul mânz al urşilor apţi	the beautiful foal$_M$ of the skilled bears$_M$
goalele nomade ale micuţei camarade	the underdressed nomads$_F$ of the small comrade$_F$
profesorul viteaz al hoţilor smezi	the brave teacher$_M$ of the pale thieves$_M$
sătulii studenţi ai hoaţei înalte	the satiated students$_M$ of the tall thief$_F$
mireasa înaltă a flămânzilor profesori	the tall bride of the hungry teachers$_M$
moartele verişoare ale frumoaselor profesoare	the dead cousins$_F$ of the beautiful teachers$_F$

Assignment 1: (6 marks)

Translate to English:

(a) hoţul flămând al copilelor viteze

(b) micuţa cetăţeană a mânjilor frumoşi

(c) studentele viclene ale înaltului nepot

[3] Romanian belongs to the Romance branch of the Indo-European family. It is spoken by approx. 24 million people primarily in Romania and Moldova.

[4] ş = sh in shop; ţ = ts in cats; j = s in vision; ă = a in about; â and î are vowels.
$_M$ = masculine; $_F$ = feminine.

[5] A foal is a baby horse.

Assignment 2: (6 marks)

Translate to Romanian. If multiple translations are possible, write only one:

(a) the brave $\text{cousins}_{\text{M}}$ of the dead foals_{M}

(b) the satiated thief_{F} of the underdressed nomads_{M}

(c) the curious cousin_{F} of the skilled child_{F}

Assignment 3: (4 marks)

Translate into Romanian in all possible ways:

(a) the small $\text{citizens}_{\text{M}}$ of the underdressed nomad_{M}

Assignment 4: (9 marks)

Explain how these phrases are structured in Romanian.

– Vlad A. Neasçu

Problem 3: Icari Dargwa (15 marks)

Here are some sentences in Icari Dargwa[6] and their English translations in random order:[7]

A. dul mazi ucibda	i. I caught the cat.
B. nusal tati ucibda	ii. The chief caught me.
C. nusal tati ucibdi	iii. The chief caught the cat.
D. tatil mazi čidažib	iv. The father saw the cat.
E. ul kurta čidažibdi	v. The girl saw the horse.
F. ustal du ucibda	vi. We caught the father.
G. ustal mazi ucib	vii. You_{pl} caught the father.
H. rirsil jabu čidažib	viii. You_{sg} saw the fox.

Assignment 1: (4 marks)

Determine the correct correspondences.

Assignment 2: (2 marks)

Translate into English:

(a) rirsil nusa ucibdi

(b) mazil du čidažibda

[6]Icari Dargwa is a variant of the Dargwa language, which belongs to the Northeast Caucasian (Nakh-Dagestani) family. It is spoken by approx. 497 thousand people in the Republic of Dagestan, Russia.

[7]c = ts in tsar, č = ch in chop, ž = s in vision.

Assignment 3: (6 marks)

Translate into Icari Dargwa:

(a) The father caught the girl.

(b) The horse caught me.

(c) The fox saw us.

Assignment 4: (3 marks)

Explain your solution.

– Kazune Sato

Problem 4: Epun (25 marks)

Here are some sentences in Epun[8] and their English translations:[9]

Midebiniz sminus fa vakvepiz.	Some kings kiss the walruses$_M$.
Chavetius f imnikiniz gub midep.	The cats$_M$ won't love this queen.
Tak huspat asa smin?	What have they$_M$ kissed?
Asa hahielogus chegodiniz.	Some dogs$_F$ haven't looked at them.
Guza hakakolgu gub zenadin.	This girl hasn't seen her.
Chupanibu fa pelikin.	The turtle$_M$ won't come.
Osozeu uza imnipiz.	He doesn't catch some cats$_F$.
Tak hagupat gub imnidin kakol?	What has this cat$_F$ seen?
Afa bozudin cherehelgu gub vletop.	The teacher$_F$ will write this letter.
Tak chupat f imnikin osoze?	What will the cat$_M$ catch?
Nisa osozenis fa chegodopiz.	We catch the dogs$_M$.
Af imnipiz bozudiniz havetigus.	Some teachers$_F$ have loved the cats$_F$.
Zaddiliniz hapanibus.	Some men have come.
Makohin binapu.	The boy returns.

Assignment 1: (3 marks)

Translate into English:

(a) Tak hugupat panib?

(b) Uza havetigu guza.

(c) Pelikop nisa hakakolnis.

Assignment 2: (8 marks)

Translate into Epun:

(a) The walrus$_F$ will kiss us.

(b) This man hasn't written a letter.

(c) Some turtles$_F$ don't look at a woman.

(d) The boys have loved the queens.

Assignment 3: (6 marks)

Translate into Epun in all possible ways:

(a) What do they love?

[8] Epun is an invented language, deliberately devised to contain constructions which violated universal grammatical principles.

[9] ch = ch in chop. The subscripts M and F refer to masculine and feminine, respectively.

(b) What will return?

Assignment 4: (8 marks)
Explain your solution.

– Vlad A. Neacşu (inspired by a problem created by Adina Dragomirescu)

Problem 5: Maldivian (18 marks)

Below are four lines of a poem written in the Maldivian[10] language, using the Thaana writing system. The Latin transcriptions in scrambled order are given on the right. [11]

Some words are missing from the Latin transcriptions and have been replaced by placeholders such as [1], [2], etc.

A.	ދާންމަ ހަދާ ކުލަހުން ކަމަށް ނޭދެނީ	i. ekam [1] avaha eythi geleyne
B.	ދެ ގަނޑު ތަކެއްގެ ތެރެ ކަ	ii. [2] mage [3] reethi lhen
C.	ތަކެއް ފަހަރު ތިން ގައި ހުންނަ	iii. moonu balaa loagadun [4] soora
D.	ތި ގަނޑު ތަކުގެ [5] ފަހު ލޯބި	iv. thi [5] enme fahu loabi

Assignment 1: (4 marks)

Determine the correct correspondences between lines (A-D) written in Thaana and the transcriptions (i-iv).

Assignment 2: (2 marks)

One of the words in Maldivian is written in an unexpected way. What is this word? Explain your answer.

Assignment 3: (5 marks)

Fill in the blanks [1] to [5].

Assignment 4: (3 marks)

Write the following names of islands in Thaana:

[10] Maldivian (also known as Dhivehi) is an Indic language spoken by approximately 350,000 people in the Maldives, and is the official language.

[11] lh and th are consonants.

(a) finey

(b) baarah

(c) angaagau

Assignment 5: (4 marks)

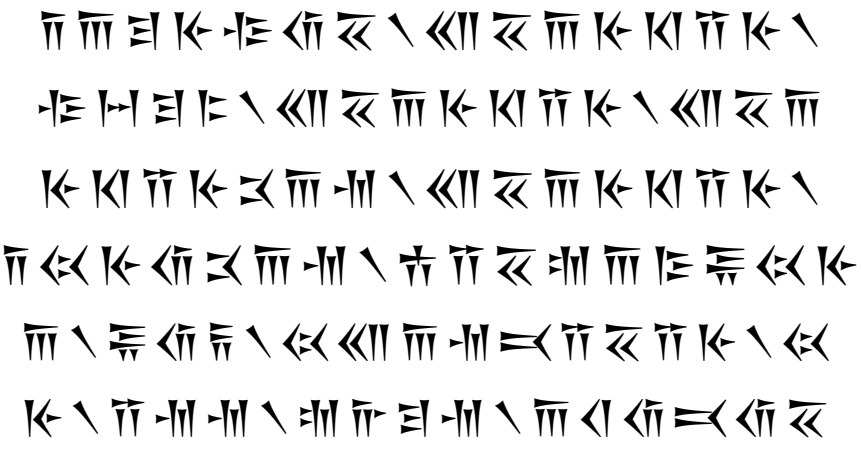
Explain how the Thaana writing system works. You do not need to list the meaning of every symbol.

– Vlad A. Neacşu

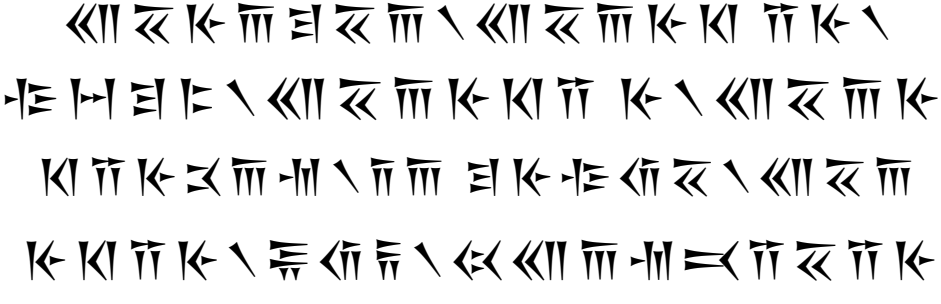
Problem 5: Old Persian (20 marks)

Old Persian was spoken from 600-300 BCE in Achaemenid Persia. It was written in cuneiform[10]. Since the medieval times travelers had noticed the Old Persian inscriptions in cuneiform carved in the palaces and ruins of Persepolis, a city in Iran. However, the script remained undeciphered till 18th century. In 1802, Georg Friedrich Grotefend, a German philologist, managed to (almost correctly) decipher most of the script starting from two simple inscriptions, known as the Niebuhr inscriptions 1 and 2.[11]

Niebuhr inscription 1:



Niebuhr Inscription 2:



Based on the repetition patterns in the two inscriptions and context in which they occur, Grotefend correctly guessed that the translation of the first inscription should be:

[10] Cuneiform was a writing system produced using wedge-shaped marks in clay tablets, inherited from the Sumerian Empire.

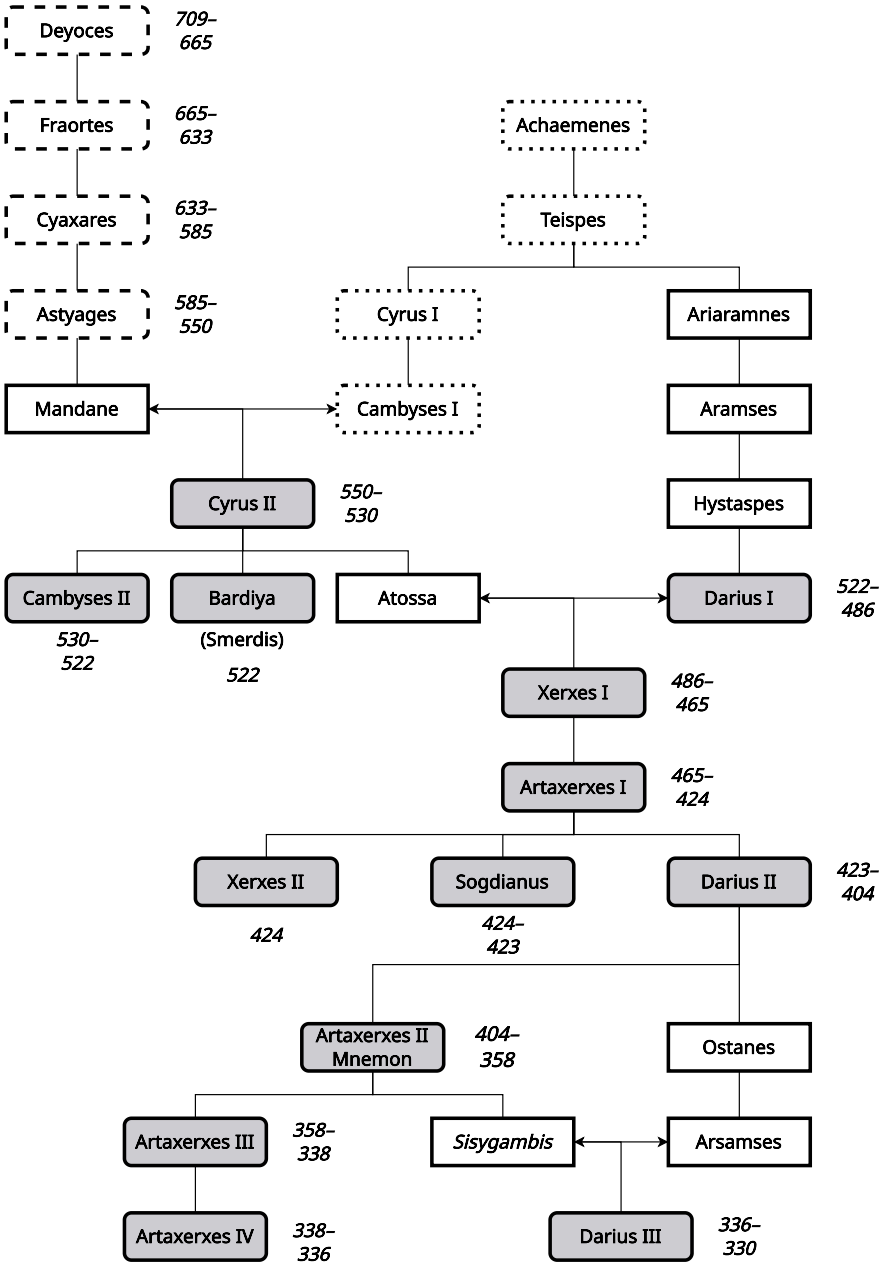
[11] Slightly simplified for this problem.

Translation 1: *A, the Great King, King of Kings, King of countries, son of B, an Achaemenian, who built this Palace.*

Note: *Achaemenian* refers to a descendant of Achaemenes.

Grotefend deciphered the above words without deciphering the letters. He could see that the second inscription also mentions two names, let's say, C and D (in that order).

At first, Grotefend was unable to decipher the names - A, B, C and D. However, the family tree of the Achaemenid dynasty gave him enough clues to pin down the names. The tree is shown below. The names follow the modern English spellings that closely reproduce the Greek pronunciations of the Old Persian words. The Achaemenid kings (descendants of Achaemenes who also reigned) are shown in shaded boxes. The reigning periods of all the kings have been shown as well.



Assignment 1: (5 marks)

Translate Niebuhr inscription 2 into English (you can use C and D for the names).

Assignment 2: (7 marks)

Identify A, B, C, and D. Explain how you arrived at your answer.

Assignment 3: (5 marks)

Give the Old Persian cuneiforms for the following English words:

(a) King

(b) Son

(c) Achaemenian

Assignment 4: (3 marks)

What do the following cuneiform symbols stand for:

(a) 

(b) 

(c) 

– Monojit Choudhury